Beyond the Number

A Diagnostic Layer That Records How Learners Answer, Not Just Whether They Were Right

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Abstract. Two learners can earn the same mark for opposite reasons: one understood the material and misread a question, the other guessed and was lucky. A single number hides this difference, so both are poorly served: the first receives no credit and no targeted help, the second receives false reassurance. This report proposes a lightweight diagnostic layer that sits on top of ordinary multiple-choice assessment. It collects three extra signals per question (a confidence rating, a misconception-coded wrong answer, and a reworded probe question after an error) and converts them into a learner profile (secure, lucky, careless, or misconceived) for each topic. We describe the design, a data model, an ML component for scaling it, and a pilot protocol. As a proof of logic, we run a synthetic simulation in which three learner types have nearly identical average scores (67.4%, 67.0%, 65.1%). A classifier using score alone identifies the learner type with 37.1% accuracy (chance is 33.3%); adding confidence raises this to 71.0%; adding the probe raises it to 89.7%. The simulation uses assumed parameters and is not evidence about real students; the real test is the pilot proposed in Section 7.

Keywords: formative assessment, confidence-based marking, misconception diagnosis, learning analytics, educational data mining

1. Introduction

Marks, grades, ticks, streaks and rankings are how learning is usually counted. They are cheap to produce, easy to compare, and say almost nothing about how a learner arrived at an answer. The consequence is a measurement gap. The number records an outcome, while the decisions that follow (what to revise, who needs help, what to teach next) depend on the process behind it.

Consider a 30-question test in which both Asha and Ravi score 20/30. Asha understood nearly everything but misread the wording of several questions. Ravi understood about two thirds and guessed the rest successfully. Their needs are far apart: Asha needs practice in careful reading and exam technique, Ravi needs to learn the content. Treated identically, Asha is under-credited and Ravi moves on with gaps that will surface in the next course.

This report addresses two questions: **(i)** what does a learner understand that a number cannot show, and **(ii)** who is let down when the number is all we look at? We answer them with a practical system rather than a new theory, and we state plainly what is and is not yet demonstrated.

2. Problem Analysis

2.1 What a single score hides

- **Luck versus knowledge.** On a four-option question, a pure guess is right 25% of the time.
- **Execution errors versus concept errors.** A misread question and a missing concept both produce a wrong answer.

- **Misconceptions.** A confidently held wrong belief is more dangerous than an acknowledged gap, because the learner does not know to fix it.
- **Calibration.** Whether the learner knows what they know is itself a learning outcome, and a score cannot reveal it.

2.2 Who is let down

Group	What the number does to them
Learner who understood but slipped	Loses credit; receives generic advice ("study more") that does not match the real problem.
Learner who guessed correctly	Receives false reassurance; the gap stays hidden until it compounds.
Learner with a confident misconception	Practises the wrong idea; mistakes feel like successes.
Teacher	Sees an average, not that 40% of the class share the same specific misunderstanding.
Institution	Rankings and streaks reward volume and luck, and can push learners to optimise the metric instead of understanding.

3. Related Work

The ideas combined here each have an established basis; the contribution of this report is their integration into one low-friction layer and a plan to test it.

Certainty-based marking. Gardner-Medwin and colleagues developed confidence-based marking in which learners state certainty and are rewarded for calibrated answers [1]. **Two-tier diagnostic tests.** Treagust proposed pairing a content question with a reasoning question whose options encode known misconceptions [2], and concept inventories such as the Force Concept Inventory use misconception-coded distractors [3]. **Procedural bug models.** Brown and Burton modelled systematic student errors as "bugs" in procedures [4]. **Knowledge tracing.** Bayesian Knowledge Tracing models guess and slip probabilities per skill [5], and deep knowledge tracing uses recurrent networks for the same purpose [6]. **Feedback research.** Black and Wiliam [7] and Hattie and Timperley [8] argue that formative, specific feedback has large effects on learning. **Miscalibration.** Kruger and Dunning [9] report that low performers often overestimate their ability, which supports measuring confidence directly.

The gap this report targets is practical. Certainty-based marking needs learner buy-in, two-tier tests need expert-written items, and knowledge tracing infers guess and slip statistically without ever asking the learner. A system that combines cheap self-report, targeted probes and AI-assisted item authoring could be adopted in an ordinary first-year course.

4. Proposed Solution

The system adds three signals to every question and one output layer. It does not replace the marks; it explains them.

4.1 Signals collected

Signal	How it is collected	What it separates
Confidence tag	After answering, tap Sure / Unsure / Guessing (one tap, under 2 seconds).	Knowledge from luck; confident errors from honest gaps.

Signal	How it is collected	What it separates
Misconception-coded options	Each wrong option is written to match a specific, named error.	Which misconception, not merely that the answer was wrong.
Reworded probe	After a wrong answer, a near-identical question with different wording and surface features.	Misreading or slip (passes the probe) from a concept gap (fails again).
Response time (optional)	Time to first answer, compared with the learner's own baseline.	Fluent recall from slow reasoning; a noisy check on the confidence tag.

4.2 The diagnostic matrix

Correctness and confidence together give four cells with different instructional meaning:

	Sure	Unsure / Guessing
Correct	**Secure.** Knows it. Move on.	**Lucky or fragile.** Right answer, weak foundation. Reinforce.
Wrong	**Danger zone.** Misconception (if the probe also fails) or careless misread (if the probe passes).	**Known gap.** The learner is aware; give direct instruction.

4.3 Learner profile, classification rules and outputs

For each topic the system aggregates the cells into a profile. A simple rule-based version already works without any ML:

- **Secure:** high share of correct-and-sure answers.
- **Lucky/guessing:** many correct answers tagged Unsure or Guessing, and probes on those items fail.
- **Careless:** wrong-and-sure answers whose probes are mostly passed.
- **Misconceived:** wrong-and-sure answers whose probes are failed, with one dominant distractor pattern.

The learner sees a profile such as *"solid on loops, lucky on recursion, misconception about pass-by-reference"* with targeted practice, rather than "70%". The teacher sees a class-level heatmap of misconceptions, ranked by how many learners hold each one. The final grade can still be reported as usual; the profile is formative and is never used to rank learners.

5. System Design and ML Component

5.1 Architecture

Stage	Component	Notes
1	Question bank with misconception codes	Each item stores the skill tag and a misconception code per distractor.
2	Response capture	Logs answer, confidence, time, probe result per attempt.
3	Diagnostic engine	Rules first, learned model second (below).
4	Learner view and teacher dashboard	Per-topic profile, next practice items, class misconception heatmap.

5.2 Where machine learning helps

- **Distractor authoring.** The costliest part of the approach is writing misconception-coded wrong options. A language model can propose candidates ("a learner who forgets the sign change would answer X") that a teacher approves or edits. Human review stays in the loop, because plausible but wrong misconceptions would corrupt the diagnosis.
- **Profile classification.** Replace hand-set thresholds with a classifier over features (share of correct-and-sure, wrong-and-sure, probe pass rate, time ratio). Start with logistic regression or gradient-boosted trees for interpretability.
- **Misconception clustering.** Cluster wrong-answer patterns across learners to discover misconceptions the teacher did not anticipate, then label them manually.
- **Calibration.** Track each learner's confidence versus accuracy over time (a calibration curve) and use it to weight their tags, which reduces the effect of habitual over- or under-confidence.

5.3 Data model (minimal)

```
response(learner_id, item_id, skill_id, chosen_option, correct, confidence, time_ms, is_probe, parent_item_id)
item(item_id, skill_id, stem, options[], distractor_misconception_code[])
profile(learner_id, skill_id, state, evidence_count, updated_at)
```

6. Proof-of-Logic Simulation

Purpose and limits. This experiment checks whether the extra signals carry information that the score lacks, under assumptions we state openly. It uses synthetic learners, so it demonstrates the logic of the design, not its effect on real students.

6.1 Setup

Each synthetic learner answers 30 four-option questions. For each question the learner either knows the material (probability k) or does not. A learner who knows may still slip (misread) with probability s. A learner who does not know answers correctly with probability g. Confidence and probe outcomes follow from whether the learner truly knew the item. Three archetypes were set to produce similar average scores:

Archetype	P(knows)	P(slip)	P(correct | not known)	P(sure | not known)	Mean score
Careless (understands, misreads)	0.85	0.25	0.25	0.20	67.4%
Guesser (gaps, lucky)	0.60	0.05	0.25	0.20	67.0%
Misconception (confidently wrong)	0.65	0.05	0.10	0.75	65.1%

A learner knowing an item is sure with probability 0.85 (0.70 if they slipped). The reworded probe is answered correctly with probability 0.90 if the item was known and with the learner's not-known success rate otherwise. Each experiment uses 1,800 learners (600 per archetype) with a 70/30 train/test split, repeated 20 times with different random seeds. A nearest-centroid classifier on standardised features tries to recover the archetype from three feature sets.

6.2 Results

Features available	Accuracy (mean ± sd over 20 runs)
Score only	37.1% ± 2.1
Score + confidence tags	71.0% ± 2.4
Score + confidence + reworded probe	89.7% ± 1.3

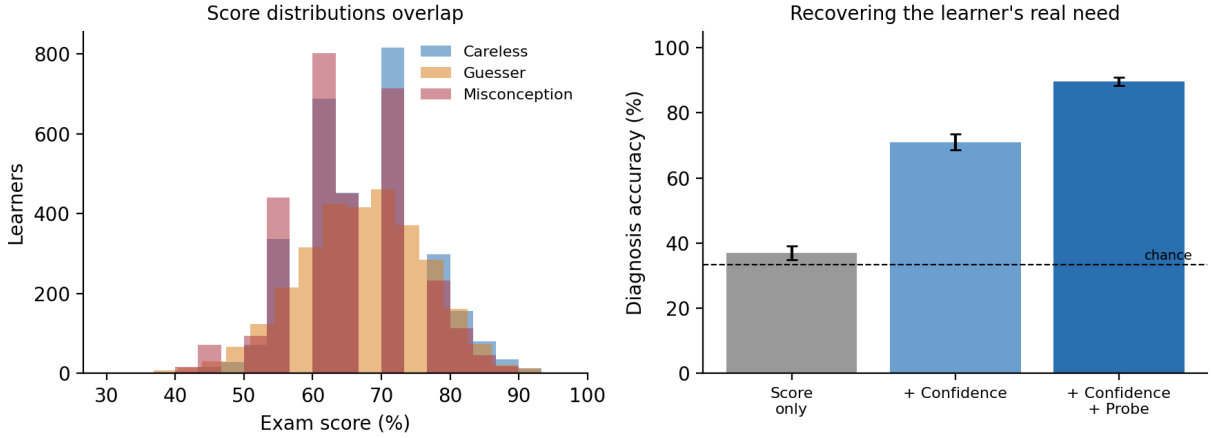


Figure 1. *Left: score distributions of the three archetypes overlap almost completely. Right: diagnosis accuracy as signals are added (dashed line = chance, 33.3%).*

6.3 Interpretation

Score alone performs near chance, as the riddle predicts: the three learners are indistinguishable by their marks. Confidence tags recover much of the difference (mainly separating guessers from the confident groups), and the probe adds the remaining step by separating careless learners from those with true misconceptions, since both are wrong-and-sure. The ordering of the three results follows directly from the assumptions, so the exact percentages should not be quoted as expected real-world performance. What the simulation does support is the structural claim: each added signal carries information that the one before it lacks.

7. Evaluation Plan for a Real Pilot

The following protocol would test the claims on real learners:

- **Setting.** One first-year course (programming or calculus), two cohorts or sections, 6 weeks of weekly quizzes.
- **Design.** Control group receives marks only. Treatment group receives marks plus the diagnostic profile and targeted practice.
- **Primary outcome.** Gain on a common end-of-unit assessment, with items different from those in the quizzes.
- **Secondary outcomes.** Calibration error (confidence versus accuracy), share of misconceptions resolved between consecutive quizzes, teacher time to identify class-wide problems, learner-reported usefulness.
- **Validity checks.** Compare profile labels with a teacher's independent judgement for a sample of learners, and with short think-aloud interviews for a smaller sample.
- **Analysis.** Report effect size with confidence intervals; check whether gains hold for lower-scoring learners, since they are the most likely to be miscalibrated.

8. Limitations, Risks and Ethics

- **Self-report is noisy.** Learners may rate confidence strategically or inconsistently. Mitigation: keep the tags formative (not graded), cross-check with probes and timing, and track calibration over time.
- **Authoring cost.** Misconception-coded distractors need subject expertise. Mitigation: AI-proposed candidates with teacher approval, and starting with the few most common misconceptions per topic.
- **Probe design.** A poor probe (too easy, too different) gives misleading diagnoses. Probes need pilot testing against item difficulty.
- **Labelling harm.** Fixed labels such as "guesser" can stigmatise. Profiles should describe the state of a skill at a moment ("fragile on recursion"), never the person, and should not be shown to others without the learner's awareness.
- **Gaming and anxiety.** Confidence tagging could add stress or be gamed. A small pilot should measure both.
- **Scope.** The method suits closed-form questions. Open-ended and project work need different evidence, such as rubrics and process logs.
- **Data protection.** Response-level data is sensitive. Collect the minimum, keep it within the institution, and give learners access to their own data.

9. Conclusion

A single mark compresses a learning process into one figure and discards the information that would tell a teacher or learner what to do next. Adding confidence tags, misconception-coded options and reworded probes recovers much of that information at a low cost to the learner. A synthetic simulation shows the logic holds: three learner types with nearly equal scores become separable once these signals are collected. Whether the approach improves learning for real students remains an empirical question, and Section 7 sets out how to answer it. The recommended first step is a narrow pilot in one course, one topic area and one classroom, rather than an attempt to fix assessment in general.

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Note: these are cited from the author's background knowledge. Verify each (authors, year, venue, page numbers) before submission.

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